



Research paper

Towards understanding the behavior of image-based network intrusion detection systems

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ABSTRACT

Network Intrusion Detection Systems play a pivotal role in preventing cyber attacks by identifying threats within computer networks. Recent advancements in deep learning techniques positioned them as highly effective methods in detecting a diverse range of cyber attacks. However, the "Black-Box" nature of deep models makes understanding their decisions very challenging, and renders them susceptible to adversarial attacks. In this paper, we propose the use of Explainable AI (XAI) approaches in deep-learning-based network traffic classifiers to validate their decisions' rationale and soundness. In particular, we combine the popular Grad-CAM technique with a reverse lookup algorithm to explain models trained using image-transformed raw network traffic sessions, encompassing general, malware, and encrypted traffic data. Model behaviors were analyzed by mapping the highly impacting pixels to their corresponding raw features, to facilitate investigating the meaningfulness of the features learned by the model. Experimental results indicate cases of consistent highlighting of pixels associated with network layers across specific traffic types. However, models occasionally used unexpected features during the classification process, raising security vulnerability concerns that merit serious investigation. The proposed approach serves as a valid method to explain the behavior of general black-box image-based network traffic classification models and assess their robustness. The implementation code is available at <https://github.com/ayahdev/XAI-Image-Based-IDS>.

1. Introduction

Network traffic classification approaches are necessary for the operation and security of network services. They are used for various purposes, including Quality of Service (QoS), Network Intrusion Detection Systems (NIDS), and resource provisioning to name a few. NIDS plays an important role in maintaining the security and integrity of computer networks by detecting various types of attacks. Other NIDS tasks also include real-time detection of potential security threats, and quick response to potential security breaches (Hnamte and Hussain, 2023).

The literature shows that numerous IDSs have been developed and deployed in network attack detection (Yi et al., 2023) using a variety of techniques including statistical methods, Machine Learning (ML) and Deep Learning (DL) algorithms. The latter methods which are extensively reviewed in Gamage and Samarabandu (2020) and Lee et al. (2021) have become widely used due to their high detection rate (Neupane et al., 2022). In particular, deep computer vision approaches

experienced significant improvements over existing traditional models, which motivated their use in NIDS applications (Al-Qatf et al., 2018; Li et al., 2017).

Among DL models, Convolutional Neural Networks (CNN) garnered much attention due to their exceptionally high classification accuracy and ability to self-learn important features from raw data (Valueva et al., 2020); also known as representation learning capability (Salman et al., 2021). In addition to their superior classification accuracy, they are known to have fast inference times (Masum and Shahriar, 2020), a desired property to handle large amounts of real-time network traffic data (Salman et al., 2021). To leverage their benefits, several NIDS researchers formulated the problem of intrusion detection as a binary image classification problem where CNNs can be used.

Although CNNs are designed for image classification, they are widely utilized for NIDS by transforming network traffic into structured image representations. This transformation enables CNNs to leverage their strengths in spatial feature extraction, allowing them to capture

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complex relationships within network traffic that may be lost in traditional packet or flow-based representation. CNN-based NIDS were demonstrated to be efficient, scalable, have high accuracy, and have low processing times (Ghadermazi et al., 2024; Bhuyan et al., 2024; Deshmukh and Ravulakollu, 2024). Techniques like depth-wise separable convolutions and model pruning further enhance their efficiency. Additionally, methods like PCA, t-SNE, and hyperparameter tuning also contribute to minimizing computational overhead (Deshmukh and Ravulakollu, 2024).

To this end, several transformation-based approaches were proposed (Li et al., 2017; Wang et al., 2017b,a; Hu et al., 2023; Guo et al., 2020), achieving good results by transforming the first several packets of a network flow/session to an $(N \times N)$ grayscale image. Certainly, network session data can reflect the correlation of network packets well, and in particular, identify traffic patterns that indicate malicious activity (Miyamoto et al., 2023). As such they are favorably used by packet-based approaches to unveil malicious network data by inspecting the packet payload (Farrukh et al., 2022). In particular, recent studies (Yujie et al., 2020; Andresini et al., 2021) compared image-based network traffic data representation to other representations and other state-of-the-art detection models respectively, and reported that this representation was superior. For instance, the study in Andresini et al. (2021) found out that an image-based model dubbed “MAG-NETO” yields an accuracy of 98.49% and F1-score of 96.28%, whereas a non-image-based Deep Neural Network model dubbed “AIDA” yields an accuracy of 94.50% and F1-score of 85.80%, when both are trained on the CICIDS-2017 dataset. Moreover, image-based representation provides a higher capacity for capturing complex patterns in the raw data and might help users identify abnormal traffic patterns more intuitively, as recently analyzed by Jeong et al. (2023).

Despite their success, CNN’s decision behavior is rarely explained in the NIDS domain and is rather treated like a black-box (Zebin et al., 2022). Explaining the behavior of CNN classifiers is particularly important in critical cases such as NIDS applications. Limited per-packet explanations were done in Garcia et al. (2022) and Cherepanov et al. (2022). However, CNNs employed in *session-based* NIDS approaches are becoming popular (Sun et al., 2023; Miyamoto et al., 2023), yet to the best of our knowledge, no works investigated the explainability of the decision behavior of those models.

Due to the dramatic increase in traffic encryption, reaching more than 80% of the global traffic (Wang et al., 2022), attackers also leverage encryption to evade detection of their malicious activities. As such, the conventional methods of classification of traffic are facing new challenges (Smadi et al., 2021). However, researchers have recently published DL approaches (Sun et al., 2023; Lotfollahi et al., 2017; Salman et al., 2021; Cao et al., 2022) claiming to be capable of finding patterns in encrypted network traffic payloads and classifying applications based on these patterns. However, according to Garcia et al. (2022), even well-trained models are incapable of detecting patterns in the encrypted payloads, therefore suggesting focusing on the packet headers rather than the payload.

Having such challenges necessitates the use of Explainable AI techniques (XAI), which refers to methods in AI applications that allow human experts to understand the results and reasoning behind the predicted values (Gunning, 2017). XAI is critical for trustworthy autonomy in network management (Garcia et al., 2022; Ribeiro et al., 2016). On the one hand, by just identifying specific decisions, XAI methods aid users to comprehend models’ rationale and assist them in their effective use and maintenance. On the other hand, transparency about predictions contributes to the development of trust in a system’s intended behavior and provides users with confidence that it performs tasks correctly (Neupane et al., 2022). However, exposing the transparency might also be exploited by attackers, which may pose severe security threats (Taddeo et al., 2019).

In this work, we target the problem of explaining CNN models trained on image-transformed traffic data to understand their behavior

and discover potential vulnerabilities. Specifically, we focus on the general interpretability of network traffic, rather than the in-depth theoretical analysis of feature interactions. We leverage a reverse look-up algorithm to understand the considered parts of the network traffic (layer and layer content) in general which, unlike natural image classification domains, is possible because the pixels here represent bytes from standard network communication protocols.

To this end, we employ the Grad-CAM technique (Selvaraju et al., 2017), a method used to highlight impactful pixels used by a CNN to classify network sessions. This study delves into understanding the behavior of NIDS using the 2D image representation. First, we aggregate the highly impacting layers (in percentages) across each traffic type examples to identify the layers that contributed significantly to the model decision across the traffic type. This is to establish the layers that consistently and significantly affect the classifier’s decision in each traffic type. We consider this aggregate estimation technique as a valid explanation method of the different patterns in the network traffic, unlike techniques that rely on analyzing individual sampled sessions. Since the models may learn to classify network sessions based on misleading features, for instance MAC addresses, we also test different alterations to the Ethernet layer in the network packets to study the effect of this layer on the models’ decisions. To this end, we investigate the use of XAI for image-based traffic classification models by focusing on the following research questions related to image-based NIDS:

- RQ1: Which network layers and features do CNN-based classifiers consider?
- RQ2: Are these features consistent across the same type of traffic?
- RQ3: Do models always look at relevant and meaningful features?
- RQ4: What are the consequences of training classifiers on raw network traffic?
- RQ5: How early can a model detect an attack in a network session?

This work paves the way to build explainable and trustworthy NIDS. In particular, we consider session-based traffic as the base for exploring the explainability of CNN models using an image datasets. Moreover, we study the effect of the masking and the removal of the Ethernet layer; the so-called feature variations on the models’ decision-making. Following this, we identify the network layers which could be easily comprised by attackers to potentially evade NIDS detection. Finally, we provide a comprehensive method to test the CNN models for early attack detection and identification. The rest of this paper is organized as follows: a summary of the related works is given in Section 2. The methodology used is detailed in Section 3. The experimental setup and results are provided in Sections 4 and 5. Finally, the discussion and conclusion are presented in Sections 6 and 7.

2. Background and related work

In recent years, the explainability of black-box IDS models in network traffic data has received increasing attention. In this section, we (i) outline the data representation of network traffic, (ii) discuss the XAI techniques used for image-based classification models, and (iii) review explainability techniques for different types of network traffic classification methods.

2.1. Network traffic data representation

Processing network traffic for ML tasks can be more complicated than processing fixed data or images. This can be attributed to several reasons such as the nested structure and variable size of network packets (Cherepanov et al., 2022). There are two main data representation methods for network traffic; raw packet-based representation and flow-based representation. Both representations can either be transformed into images or left as raw data. The former encapsulates the entire packet payload besides the header, while flow-based data represents the

aggregated information of related packets of network traffic in the form of flows (Andreas et al., 2020). Many researchers use the flow-based representation of network traffic. In Patil et al. (2022), researchers extract 10 out of the 76 features provided in their dataset for training purposes, while in Caforio et al. (2021), the features extracted from the flows are converted to images. Alternatively, Sun et al. (2023) introduced a new concept called “Packet Blocks”. Since a packet, in most studies, is regarded as the basic unit of a flow, the researchers group continuous packets of a flow that move in the same direction and name them Packet Blocks. The flows are further divided into Sessions, and features are extracted from the Packet Blocks of the sessions to be displayed in 2-dimensional images. These images are then used as the input for their model.

Other works, including Cherepanov et al. (2022) and Bierbrauer et al. (2023), use the raw packet data to build their classification models. In Bierbrauer et al. (2023), the authors extract data from the network packets’ TCP and UDP payloads. The number of bytes in the protocols is limited to 1500 bytes and are used as input vectors to the model. The researchers in Bierbrauer et al. (2023) extract specific data from the network packets in Cherepanov et al. (2022), then clean and pre-process full network packets for the classification. The small network packets without payloads are discarded, and the TCP/UDP headers are padded to be the same length. The results are then converted into image representations of the network packets for training, and therefore, they are trimmed/padded to be of a fixed length of 1480 bytes.

Two recent studies, Li et al. (2022) and Salman et al. (2021), utilized both flow and raw network traffic representations. In Li et al. (2022), 784 bytes were extracted from the flow and session-based representations, and converted to grayscale images of fixed size to be used as inputs for the model. In Salman et al. (2021), the authors performed a comparative study between the two data representations. The Pcap files were filtered to keep the TCP and UDP traffic only. Flows were then extracted and pre-processed, and further divided into sub-flows. The packet-based representation considers the first packet of each flow as raw data, and is displayed as a grayscale ($N \times N$) image. For the flow-based representation, four features were extracted from the first ($N \times N$) packets of a flow, and were displayed as an RGBA ($N \times N \times 4$) image.

Explainable AI methods such as Grad-CAM and LIME have been widely used in recent studies. Both Grad-CAM and LIME provide local explanations on a per-example basis. In Grad-CAM, which builds on the Class Activation Map (CAM) technique (Zhou et al., 2015), the gradients of the target class are computed with respect to the feature maps of a convolutional layer. These gradients are then backpropagated and globally averaged across all spatial locations in the feature map to obtain a single representative importance value per feature map. This process determines the importance of each feature map for the target class (Zhou et al., 2015) by generating a heatmap that highlights the regions of an input image that most significantly impact the model’s prediction. Although Grad-CAM inherently provides local explanations, it can also offer insights into the model’s global behavior when applied systematically across multiple samples and analyzed using specific metrics such as Mean Observed Dissimilarity (MOD) and Variation in Dissimilarity (VID) (Chakraborty et al., 2022).

2.2. XAI for image-based models

Many methods for explaining black-box models on a wide scale have been recently proposed. For instance numeric explanations for data mining and ML (Gilpin et al., 2018), visual explanations for visualization and diagnosis of CNN (Zhang and Zhu, 2018), and textual explanations (Barratt, 2017) have been described and grouped by the formats of the explanations generated by the models (Vilone and Longo, 2021). A very recent survey (Capuano et al., 2022) explored XAI for cyber security, while also marking warnings towards

XAI since it may also be explainable to the attacker, which may pose severe security threats (Taddeo et al., 2019). This survey distinguishes between Interpretability and Explainability where Interpretability is concerned with understanding the cause and effect within an AI system. However, explainability goes beyond interpretability by providing a human-readable explanation of how and why a model produced a certain prediction (Capuano et al., 2022). Both the Explainability and Interpretability of ML/DL models are critical for trustworthy autonomy in network management (Garcia et al., 2022).

Next, we consider related works that use primary post-hoc methods which explain a model’s behavior by using an external explainer at testing time. The external explainer uses the final convolutional layer of the CNN model to generate a heatmap highlighting the influential regions in the image for predicting a certain class without affecting the model’s performance. There are many post-hoc methods that provide local and global explanations such as Grad-CAM (Selvaraju et al., 2017), SHAP (Lundberg and Lee, 2017) and LIME (Ribeiro et al., 2016), to name a few, and a detailed review is addressed in Capuano et al. (2022). In the case of images, each pixel is considered an input as is each byte of a packet (Wang et al., 2017c), therefore, it is also possible to apply techniques like Class Activation Mapping (CAM) to show the impact of bytes in network traffic Pcap files.

2.3. XAI for traffic classification

In Patil et al. (2022), the authors utilized LIME to explain the model’s decision on the features extracted from the flow-based network traffic. Similarly, the authors in Caforio et al. (2021) applied Grad-CAM to create gradient-based visual explanations of the CNN binary classification of network flow traces represented by images. The main contribution of this work is the combination of Nearest-Neighbor search with Grad-CAM explanations to raise the transparency of network traffic CNN black-box models and enhance the post-hoc accuracy of their decisions. A recent work (Andresini et al., 2022) for traffic flow classification uses attention coupled with a multi-output DL strategy. The idea helps to better discriminate between images converted from flow characteristics of network traffic data. The authors considered visual explanation maps produced through Grad-CAM as post-hoc explanations. However, the visual explanation maps produced through attention always achieve a higher ability to make intrinsic factors for classification compared to Grad-CAM maps. The work by Galli et al. (2021) analyzed the impact of adversary attacks on XAI methods using two perturbation attacks (IFGSM and DeepFool) on four CNNs to generate adversarial samples. Two methods of XAI (layered Grad-CAM and guided Grad-CAM) were used to detect adversarial attacks. The authors demonstrated that, to an individual unaware of the attack, the attack left significant changes on the interpretations generated by XAI. They concluded that explainability of results is not an accurate method to detect adversarial attacks.

The authors in Liu et al. (2021) presented a framework to enhance the use of AI for detecting intrusions by utilizing data cleaning methods. Specifically, data cleaning and XAI techniques were employed to enhance the explainability of the intrusion detection alerts. The results demonstrated that the use of XAI and data-cleaning methods provided quality explanations for the analysts.

Targeting the explanation of image-based models trained on encrypted traffic, an interesting study conducted by Cherepanov et al. (2022) combined traffic classification of raw network data with a classification model explanation technique. They applied CAM in a CNN model to classify packets sent over a network and explain the model’s prediction. Finally, they show CAMs to the expert for further investigation. An important finding is that when classifying encrypted data, only five application classes had the impacting bytes from the beginning of the payload, namely Spotify, Vimeo, ICQ, SCP, and VoipBuster. These bytes likely represent the unencrypted headers of these applications or higher-level protocols. In most other cases, their model makes decisions

on the payload bytes of the packets and only learns the properties from the headers. With the provided explainability of model decisions through CAMs, it is observable that models do not make decisions on the encrypted parts of the packets as claimed in Lotfollahi et al. (2017). To investigate whether our method achieves the goal of Cherepanov et al. (2022), we conducted an Ethernet layer ablation study by manipulating the Ethernet layer's content using two distinct techniques as described in Section 3. The findings from our study also indicate that models tend to focus on unencrypted headers and payloads for classification, rather than encrypted parts as claimed in Lotfollahi et al. (2017).

A representative work is given by Garcia et al. (2022), where the authors attempted to interpret a black-box DL model for encrypted network traffic classification. They used an open-source interpretability toolbox to implement saliency attribution for post-hoc explanations. For visualization, they used WireShark and Scapy as a pane of hexadecimal bytes, with rows sixteen bytes long. Their results show that the model focuses on dominant features residing in the packets header portion with minor other input features combinations to classify each given task. Moreover, they optimized the training pipeline while significantly reducing the size of the DL model. Similarly, the conclusions from our aforementioned ablation study align with the work in Garcia et al. (2022), as we also observed that our models primarily relied on the header in network data, and the use of interpretability methods like Grad-CAM helped us reveal where the model focused.

All these related works exploited DL models to learn about the patterns in network traffic without human effort. They succeeded in understanding such unstructured data, differentiating between classes and identifying complex patterns between them. Most of them aimed to explain the model's decision as human-interpretable network traffic features. Given that DL models as black-box methods are becoming increasingly ubiquitous, support methods are needed to explain such models to build trustworthy NIDS.

3. Methodology

This section provides an overview of the experimental setup for the image-based DL method while highlighting the trustworthiness issues we wish to explain. We show how explainability can aid in understanding two main test cases: (i) Attack and General Traffic Classification, and (ii) Encrypted and Non-Encrypted Traffic Classification, as shown in Fig. 1. To accomplish this task, we first created labeled image datasets as described in Section 4.2. We subsequently used these datasets to train various CNN models to serve as IDS for classifying the different types of network traffic.

In the first Test Case, we create two models that classify: (i) Benign/Malicious Traffic [2 Classes] and (ii) Malware+General Traffic [20 Classes]. In the second Test Case, models are created for classifying: (i) Encrypted Traffic [6 Classes] and (ii) Non-Encrypted Traffic [6 Classes]. Moreover, as part of our investigation on whether models always look at relevant and meaningful features, we evaluate different feature variations under each test case by altering or removing the Ethernet layer. For that purpose, in both Test Cases, we experimented with three representation variations, (i): keeping the Ethernet layer intact (W_Eth), (ii): replacing Ethernet headers with 0s (Z_Eth) similar to Garcia et al. (2022), and (iii) eliminating the Ethernet layer (No_Eth) as done in Lotfollahi et al. (2017). In all cases, we aim to better understand the Ethernet layer contribution in the decisions. The naming convention we use for each experiment is the Test Case number followed by experiment designation. For instance 1.a-W_Eth denotes Benign vs. Malicious traffic with Ethernet layer.

To set the models' hyperparameters, since we are reusing previously published NIDS model architectures, we adhered to their original hyperparameter settings as specified in their respective papers and source code, ensuring consistency with prior work wherever possible. The resulting model was then used for generating activation maps using

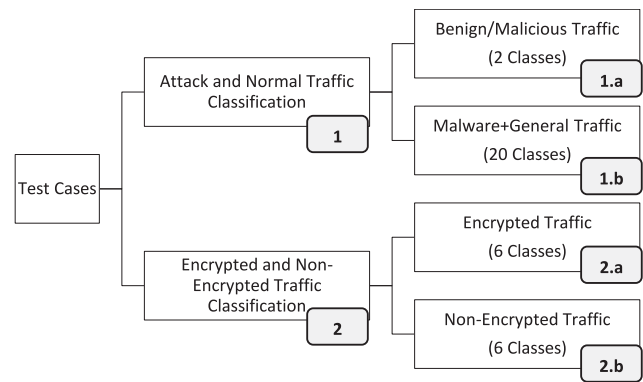


Fig. 1. Test cases designation.

Grad-CAM (Selvaraju et al., 2017). Compared to other XAI methods, Grad-CAM is computationally efficient and has intuitive visual explanations. Moreover, Grad-CAM is faster compared to SHAP (Tempel et al., 2025), requires only one forward and one backward pass (Selvaraju et al., 2017); and can be applied to various CNNs without adjusting the network structure, thus having better versatility. This makes Grad-CAM viable for real-time applications when integrated with optimized CNN-based NIDS. We explore utilizing Grad-CAM averaged activation maps for post-hoc explanations, which were initially designed to visualize the gradients of a class score for a single input image. An overview of the three main parts of our method is shown in Fig. 2.

3.1. Transforming sessions to images

The first step in the preprocessing stage is splitting the raw network traffic into sessions. We used the toolkit developed by Wang et al. (2017c) which includes four preprocessing steps; Network Traffic Splitting, Network Traffic Clearing, Image Generation, and IDX Conversion. Specifically, we used the first two tools¹ to split network traffic into sessions, clear them, and trim them to the required length for image generation.

In the splitting step, we split each Pcap file in the dataset into sessions of bidirectional flows. A flow is defined as all packets with the same 5-tuple, i.e. source IP, source port, destination IP, destination port, and transport-level protocol (Li et al., 2017; Wang et al., 2017b). The Pcap files of the sessions are then filtered and trimmed to the required length. Then, each processed Pcap file is converted to a hexadecimal string as a compact representation and reshaped into a matrix of unsigned 8-bit integers (grayscale pixel values). Next, we create a grayscale image object and save it locally, as done in Wang et al. (2017c). After analyzing the network packet layers, we chose to keep all the data from the network protocols since they may contain important traffic characteristics, as explained in Li et al. (2022). Details of the followed preprocessing steps to transform a network traffic session into its corresponding image representation are provided below:

1. Network Traffic Splitting: Pcap files are split into sessions of bidirectional flows, defined by the 5-tuple (source IP, source port, destination IP, destination port, and transport-level protocol).
2. Session Trimming and Filtering: The Pcap files of the sessions are filtered and trimmed/padded to a fixed length (1024 bytes in our study) to ensure consistency in image size.
3. Metadata Generation: The position of the start bytes of all packets from a session are saved into a JSON file to be used later in the reverse lookup algorithm described in Section 3.3.2.

¹ <https://github.com/yungshenglu/USTC-TK2016/tree/master>.

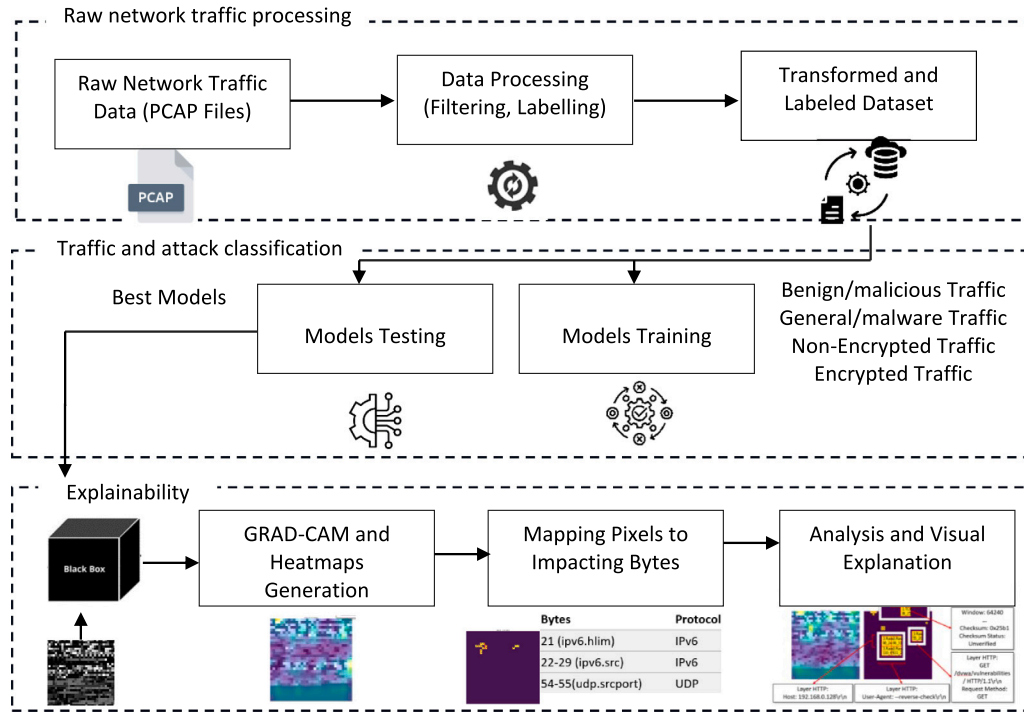


Fig. 2. Method overview.

- Hexadecimal Conversion: The processed Pcap files are converted to hexadecimal strings, which serve as a compact representation of the raw data.
- Image Generation: Each hexadecimal string is reshaped into a matrix of unsigned 8-bit integers (grayscale pixel values), creating a grayscale image. This step also involves saving the image locally.

3.2. Model structures and training

The aim of this study is to explain the decisions made by image-based network traffic classification models; CNNs are DL models designed to handle image recognition tasks (Salman et al., 2021). Our work uses 2D-CNN models to classify the grayscale images generated from the raw network traffic. Each convolution layer is followed by batch normalization and max-pooling, which is designed to reduce the complexity of the network and thus reduce training computation costs. Details of the models' structures are summarized in Tables 1 and 2. Rectified Linear Unit (ReLU) is used in all the models as an activation function, and Adamax and Adam optimization are used in Test Case 1 (i.e. Attack Traffic Classification) and Test Case 2 (i.e. Encrypted Traffic Classification) respectively. The hyperparameters are summarized in Table 3.

3.3. Applying XAI and packet reversal techniques

To understand and evaluate the trustfulness of the classification models, we use XAI to extract the features that have a high impact on the models' decisions. Our choice of XAI method is Grad-CAM. Once the models are trained, we use the images from the test splits and generate their activation maps. A reverse technique is implemented using the packet start-bytes stored in JSON files with each session image. Details of the mentioned steps are explained in the following sections.

3.3.1. Heatmap generation using Grad-CAM

In this work, we use Grad-CAM for visual explanations to understand the decisions made by the classifiers.

The input image is passed to the Grad-CAM explainer along with the trained model, which returns a heatmap that highlights the pixels influencing the decision. In our visualizations, pixels with the highest gradient values are represented with different "Yellow" colormap ranges. We extract only the most impactful yellow-colored bytes and apply the reverse lookup technique to explain the trained model's decision-making. Fig. 3(a) shows a heatmap of a sample from the class "Malicious - SQLI".

3.3.2. Reversal technique on generated activation maps

As further explained in Section 4.2, we collected the start byte of each hexadecimal packet converted to an image and stored them in a JSON file to be used later during the reverse process. By using the start bytes of the hexadecimal packets, we can collect detailed information on the positions of the Yellow bytes in the activation maps. For each yellow-highlighted byte in the heatmap, we retrieve: (i) Packet Number, (ii) Position of Hex Byte in Packet, (iii) Protocol Name, (iv) Position of Hex Byte in Protocol, and (v) Hex Byte Value.

Once this information is stored, we are now able to interpret what each Yellow Hexadecimal Byte represents in the Packet Layer Data. A Binary Mask containing the Yellow Bytes information is created for each heatmap. Fig. 3(c) shows the Binary Mask created for the heatmap in Fig. 3(a). Each Yellow Box is one Yellow Hexadecimal Byte, containing the (i) Packet Number, (ii) Protocol, (iii) Position in the Packet, and (iv) Hex Byte Value of the Yellow Hexadecimal byte.

For interpreting the Packet Layer Data that each Yellow Hexadecimal byte represents, a command that converts a Pcap file to a Packet Description Markup Language (PDML) file from "tshark" is used. The PDML file of a Pcap file contains all the Packet, Protocol, Hexadecimal, and Position data of all the packets. We map those positions acquired in the previous step with the data in the PDML file and return what each Yellow byte represents. Fig. 3(b) shows the Packet Layer Data representation of the Yellow Hexadecimal Bytes from the Heatmap and

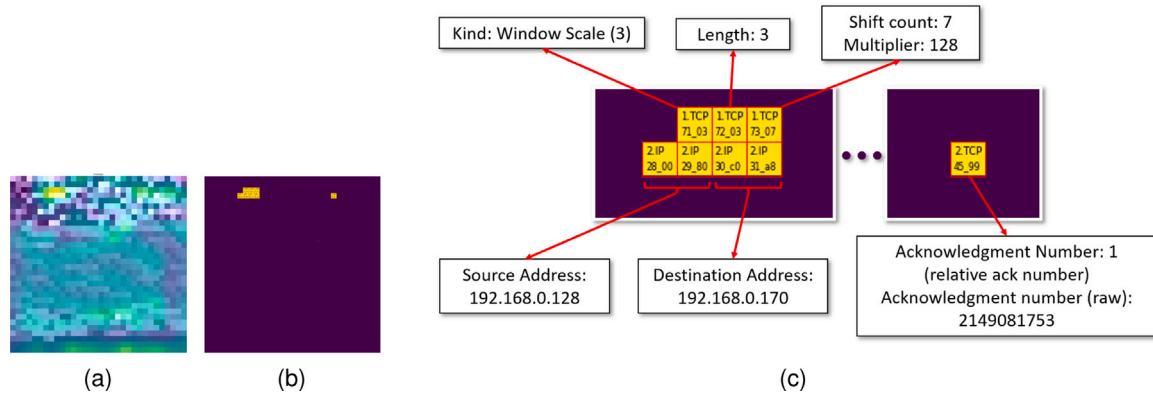


Fig. 3. (a) Heatmap, (b) Binary mask, and (c) Visual explanation of highlighted Packet Layer Data Representation of Yellow Hex Bytes.

Table 1

Structure of models in Test Case 1.

Model structure	Output shape - Binary	Output shape - Multi-class
Convolutional 2D	(32, 32, 8)	(32, 32, 32)
Batch Normalization	(32, 32, 8)	(32, 32, 32)
Max Pooling 2D	(16, 16, 8)	(16, 16, 32)
Dropout	(16, 16, 8)	(16, 16, 32)
Convolutional 2D	(16, 16, 16)	(16, 16, 64)
Batch Normalization	(16, 16, 16)	(16, 16, 64)
Max Pooling 2D	(8, 8, 16)	(8, 8, 64)
Dropout	(8, 8, 16)	(8, 8, 64)
Flatten	(1024)	(4096)
Dense	(128)	(128)
Batch Normalization	(128)	(128)
Dropout	(128)	(128)
Dense (act = "sigmoid"/"softmax")	(1)	(20)

Table 2

Structure of models in Test Case 2.

Model structure	Output shape
Convolutional 2D	(32, 32, 32)
Max Pooling 2D	(16, 16, 32)
Convolutional 2D	(16, 16, 64)
Max Pooling 2D	(8, 8, 64)
Flatten	(4096)
Dense	(1024)
Dense(act = "softmax")	(6)

Table 3

Model hyperparameters.

Test case	Epochs	LR	Batch size	K-Fold ?
1.a	50	0.001	128	Yes - 5 Folds
1.b	100	0.01	256	No
2.a	50	0.001	100	No
2.b	50	0.001	100	Yes - 5 Folds

Binary Mask in Figs. 3(a) and 3(c). Fig. 4 shows the steps conducted in the reverse lookup technique explained above.

As for the calculation of the percentage contributions, shown in Table 12, these are quantified as follows; for each heatmap, we check whether a highlighted Yellow pixel from each network protocol exists. Counters are used to keep track of the number of images with at least a single yellow pixel from each protocol. Then, the result is divided by the number of samples in each test set.

4. Experimental setup

4.1. Datasets

To test the proposed approach, we used the following public datasets that are both popular and represent a great variation of

real-world network traffic. However, these datasets do contain biases and have class imbalances that can influence the model's performance. To address class imbalance and bias, we applied down-sampling on majority classes to prevent model bias across different datasets. While down-sampling mitigated the problem for some datasets, in others, class imbalance remained an issue due to severely underrepresented classes, particularly in the encrypted/non-encrypted traffic classification task. The datasets' details are provided below:

CICIDS2017 (CIC)²: This dataset provides the labeled flow data for network intrusion detection research. The testbed includes data from two networks: one for attacks and the other for victims. The captured dataset contains several different types of attacks including DoS, Web attacks, Infiltration attacks, Botnet attacks, Port Scans, as well as background normal traffic that is generated by using the abstract behavior of 25 users based on the HTTP/HTTPS, FTP, SSH, and email protocols. The dataset includes raw Pcap files for each day.

In our work, we generate images from the raw network traffic data. We chose 5 main types of attacks to be used and labeled as the Malicious class, including Web Brute Force, Web SQL Injection, Web XSS, FTP, and SSH attacks. Tuesday and Thursday Pcap files contained both benign and malicious traffic, thus, sessions were filtered based on (i) the details of the attack timing provided by the researchers and (ii) simple tshark filtering commands.

The total number of malicious samples resulted from filtering was only 1609 samples. Therefore, more samples of the same attack types were collected from two other datasets; CSE-CICIDS2018 and Edge-IIoTset Cyber Security Dataset of IoT and IIoT which are described next.

CSE-CICIDS2018 (CSE)³: Similar to CICIDS2017, the dataset contains most of the up-to-date attacks and was created to provide a

² <https://www.unb.ca/cic/datasets/ids-2017.html>.

³ <https://www.unb.ca/cic/datasets/ids-2018.html>.

Table 4

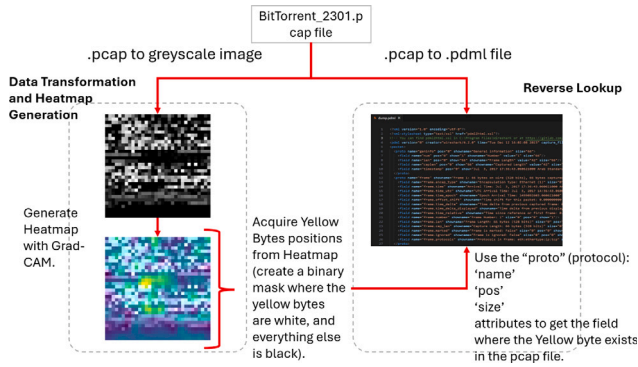
Labeled sessions from the three datasets. B and M stand for Benign and Malicious, respectively.

Dataset	Class	Traffic type count	Samples used	Train/Val/Test	% Below 1024 bytes
CIC, CSE, IOT	B	(CIC) 1	8684	63%/17%/20%	21.8%
	M	5			

Table 5

Labeled USTC-TFC2016 Multi-class sessions. N and M stand for Normal and Malware, respectively.

Dataset	Class	Traffic type count	Samples used	Train/Val/Test	% Below 1024 bytes
USTC- TFC2016	N	10	40,000	76.5%/8.5%/15%	62.4%
	M	10			

**Fig. 4.** Diagram showing the reverse lookup technique to acquire interpretation of the Yellow bytes.

benchmark dataset for IDS. The dataset contains 7 different attacks, namely: Brute-Force, Heartbleed, Botnet, DoS, DDoS, Web Attacks, and Infiltration attacks. Three attack types were used from this dataset's Malicious class, namely Web Brute Force, Web SQL Injection, and Web XSS attacks.

Edge-IIoTset (IoT)⁴: The Edge-IIoTset is a realistic cyber security dataset customized for the requirements of IoT and IIoT applications (Ferrag et al., 2022). It provides seven layers of emerging technologies generated from more than 10 different types of IoT devices. There are fourteen IoT attacks given, and they are grouped into five main categories; DoS/DDoS, Man in the Middle, Information Gathering, Injection, and Malware attacks. In our work, we use the SQL Injection and XSS attacks from this dataset to allow human interpretation. Almost half the samples (2450) were taken from this dataset for the Training/Testing process of the models.

A summary of the final culminated dataset is given in Table 4.

USTC-TFC2016⁵: The authors in Wang et al. (2017c) created this dataset to solve the issue of a lack of public raw traffic datasets. The dataset is divided into two main parts: Malware and Normal traffic, each with 10 subtraffic types as follows: Normal [BitTorrent, Facetime, FTP, Gmail, MySQL, Outlook, Skype, SMB, Weibo, WorldOfWarcraft], and Malware [Cridex, Geodo, Htbot, Miuref, Neris, Nsis-ay, Shifu, Tinba, Virut, Zeus] In our work, we use samples from all the traffic types to create a multi-class network traffic classifier. To limit prohibitive computational costs due to the large size of the dataset, only 2000 samples are taken from each traffic type. A summary of what the dataset looks like is shown in Table 5.

ISCX-VPN2016 (VPN-nonVPN)⁶: The University of New Brunswick (UNB) made the ISCX-VPN2016 (Draper-Gil et al., 2016) available for

public use. The dataset was collected by capturing experimental network traffic on a real-time basis in a customized lab. This dataset takes into account seven different types of traffic: Web Browsing, Email, Chat, Streaming, File Transfer, VoIP, and P2P, resulting in 28 GB of collected traffic data. In addition, traffic was categorized into two types based on whether applications communicated over non-encrypted or encrypted (VPN) channels. Table 6 shows the datasets distribution using 6 classes for each traffic type instead of 7, excluding the class “Web Browsing” similar to Wang et al. (2017b), due to a known problem in the labeling process.

4.2. Datasets preprocessing

After downloading the raw network traffic, dataset preprocessing was applied where needed. For CICIDS2017 and CSECICIDS2018, we used tshark filtering to obtain malicious sessions using specific regular expressions. No preprocessing was required for Edge-IIoTset and ISCX-VPN2016, as these datasets were already structured well and labeled. In USTC-TFC2016, traffic is provided as large pcap files per class, and we split them into individual sessions.

To investigate which network layers and features the CNN-based classifiers typically consider (RQ1), we trained a CNN-based NIDS model on network traffic data after being transformed into images and then applied the Grad-CAM technique to highlight the important layers and features that were considered on a set of example sessions. The process starts with preprocessing the public raw network traffic datasets for the purpose of model training and testing. Yet, the challenge is that the used datasets are raw bytes provided in Pcap files without labels (in the case of CICIDS2017 and CSE-CICIDS2018). For CICIDS2017, the contained labeled flow data are used to identify which packets correspond to malicious or benign flows. Therefore, we extended the method described in Farrukh et al. (2022) for session labeling. The output of this step is a labeled dataset of raw network sessions. As can be seen in Tables 4, 5, and 6, a significant percentage of the samples are large, and therefore, we limited the number of data bytes to include 1024 bytes (32 by 32), which captures all bytes in a network session with the Ethernet layer and header fields. The resulting 1024 bytes are converted into 32 by 32 grayscale images as done in Li et al. (2017) while also storing the start byte of each hexadecimal packet in a JSON file for each image to be used later in the reverse method. After training the models, we use the heatmaps generated with Grad-CAM on the testing sets and explain the most prominently observed behaviors in detail (RQ2). This is achieved by aggregating and ranking the network layer's contribution to the classification across various traffic types using heatmaps generated by Grad-CAM together with the reverse lookup algorithm. Additionally, we process the raw data to obtain images with the different feature variations, as explained in Section 3.

4.3. Classification model settings

Initial settings focused on the feasibility of using image-based models for raw session classification, either as attack classification or as

⁴ <https://www.kaggle.com/datasets/mohamedamineferrag/edgeiiotset-cyber-security-dataset-of-iiot-iiot>.

⁵ <https://github.com/yungshenglu/USTC-TFC2016>.

⁶ <https://www.unb.ca/cic/datasets/vpn.html>.

Table 6
Labeled VPN-nonVPN-2016 sessions.

Dataset	Class	Traffic type count	Samples used	Train/Val/Test	% Below 1024 bytes
VPN- nonVPN	Non-encrypted	6	26,921	80%/10%/10%	85%
	Encrypted	6	8579		65%

traffic type classification. There are many research works in the literature that target session classification, much of them were reported previously in [De Lucia et al. \(2021\)](#). For the first Test Case - *Attack Traffic Classification*, we implemented our own work while following common hyperparameters recommended by other papers with the datasets explained in Section 4.1. Considering the second Test Case - *Encrypted Traffic Classification*, we reproduced the work carried out in [Wang et al. \(2017b\)](#) and explained it here as it lays the groundwork for our primary findings with respect to XAI. We considered the 2D CNN architecture along with different session-based data preprocessing versions. To analyze and examine if CNN-based models consistently focus on relevant and meaningful features (RQ3), we employed 3 feature variations (mentioned in Section 3) for each task: (1) utilizing the entire packet raw data and (2) removing the Ethernet layer raw data using two distinct techniques. This is to help us understand this layer's contribution to the predictions made. To assess the consequences of training classifiers on raw network traffic (RQ4), we analyze cases where padding and trimming of the sessions might cause the models to use irrelevant or misleading features for decision-making. Finally, we generate sessions with different lengths, each having packets that range from one to a given maximum number P of packets to determine how early a model can detect an attack in a network session (RQ5).

4.4. Evaluation metrics

Generally, the performance of a classifier is evaluated through four metrics; accuracy (AC), precision (P), recall (R), and F-measure (F). These metrics are further defined by true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Specifically, TP and TN are the number of attacks and normal records predicted correctly, respectively. Conversely, FP and FN represent the number of normal and attack records that were incorrectly classified. Among all the metrics, F1 is considered an important indicator in the field of traffic classification ([Li et al., 2022](#)). During training, accuracy was monitored across the training and validation sets to ensure convergence and reduce potential overfitting.

5. Results

This work aims to explain decision-making for black-box models that are trained to classify image-transformed network traffic data. In particular, we investigate retraining techniques to evaluate network layers and feature attributions to explain the consistency of classifications and misclassifications of session-based traffic. To this end, we divided the results into three sections; with Ethernet information intact, with Ethernet replaced by 0s, and after removing Ethernet entirely. Furthermore, we investigate important issues related to the raw bytes preprocessing used in other research works and their associated consequences.

5.1. Classification performance results

In each subsection, we describe all three experiments (W_Eth, Z_Eth, and No_Eth) that were carried out and compare the results of the models after performing 5-Fold Cross Validation on each feature variation (15 classifiers in total: 5 W_Eth, 5 Z_Eth, and 5 No_Eth). The classifier with the best result from each experiment, after macro-averaging the Precision, Recall, and F1-Score of classes, was selected.

Table 7
Performance metrics for Attack Traffic classification, Test Case 1.

Measure	1.a CIC + CSE + IoT			1.b USTC-TK2016		
	W_Eth	Z_Eth	No_Eth	W_Eth	Z_Eth	No_Eth
Val_Acc	1.000	1.000	0.997	0.990	0.987	0.982
Precision	1.000	0.998	1.000	0.983	0.982	0.983
Recall	1.000	1.000	0.998	0.983	0.982	0.983
F1-Score	1.000	0.999	0.998	1.000	1.000	1.000
Accuracy	1.000	0.999	0.998	0.980	0.980	0.980

Table 8
Performance metrics for Encrypted (VPN) and Non-Encrypted Traffic (Non-VPN) classification, Test Case 2.

Measure	2.a VPN			2.b Non-VPN		
	W_Eth	Z_Eth	No_Eth	W_Eth	Z_Eth	No_Eth
Val_Acc	0.942	0.956	0.968	0.763	0.751	0.739
Precision	0.938	0.944	0.965	0.761	0.748	0.714
Recall	0.936	0.945	0.965	0.761	0.747	0.709
F1-Score	0.937	0.944	0.965	0.760	0.747	0.706
Accuracy	0.999	0.993	0.997	0.888	0.852	0.821

Test Case 1: Attack and normal traffic classification. For this Test Case, competitively good results were achieved under the dataset, which is fairly large. A summary of the results is shown in [Table 7](#). The models performed well on both test cases under different feature variations. In Test Case 1.a, the classifier W_Eth had the best performance, followed by Z_Eth. Both best performing models were trained on data which includes the Ethernet layer, while the model with an inferior performance was trained on zeroed data instead of actual data from the Ethernet layer. In Test Case 1.b, the models' performance is fairly close, varying slightly in the Validation Accuracy, Precision, and Recall. A more detailed explanation of the behavior of these models will be provided in the following section.

Test Case 2: Encrypted and non-encrypted traffic classification. After the training and testing procedure of the CNN models, we chose the best models having highest performance metrics for each feature variation. The performance results are shown in [Table 8](#). The models performed well on both the VPN and Non-VPN traffic. Surprisingly, the performance of the model was higher in encrypted traffic than non-encrypted traffic. For the latter, the highest accuracy is achieved with W_Eth (0.888), and accuracy declines as the Ethernet layer is removed, suggesting that Ethernet headers provide valuable information for non-encrypted traffic classification. Next, we leverage Grad-CAM and reverse techniques to understand the underlying reasons behind the model behavior in Section 5.2.3.

5.2. XAI and packet reversal technique results

The purpose of the next set of experiments is to explore the effect of retraining models using the three different feature variations on the model's performance. In particular, the results will include the two Test Cases, with two experiments each. The subsequent tables in this section will show each configuration tested along with its associated priority of packet layers or protocols used in descending order. Specifically, the order of the layers/protocols is based on the number of images that had at least one Yellow Hexadecimal Byte from the specific protocol or layer. Note that the leftmost protocols are most significant and rightmost contributed the least to the prediction decision.

Table 9

Impact order of layers/protocols for models in W_Eth experiments (descending order of importance). The percentage shows the portion of images having at least one Yellow pixel residing in the Ethernet layer.

Experiment	Layers/Protocols	With Ether
1.a-W_Eth	IP/Ether/TCP/Raw/DNS/UDP/Padding/LDAP/IPv6	73.92%
1.b-W_Eth	Raw/IP/TCP/Ether/DNS/UDP/Padding/IPv6	40.11%
2.a-W_Eth	IP/Ether/Raw/TCP/DNS/UDP/IPv6/ARP/Padding	46.62%
2.b-W_Eth	DNS/IP/TCP/Raw/UDP/IPv6/Ether/NTP	0.16%

Table 10

Impact order of layers/protocols for models in Z_Eth experiments (descending order of importance). The percentage shows the portion of images having at least one Yellow pixel residing in the Ethernet layer that is now represented as 0s.

Experiment	Layers/Protocols	Empty Ether
1.a-Z_Eth	IP/TCP/Raw/DNS/Ether/UDP/Padding/LDAP	15.8%
1.b-Z_Eth	Raw/IP/TCP/Ether/DNS/Padding/UDP/IPv6	20.38%
2.a-Z_Eth	IP/Ether/Raw/IPv6/TCP/DNS/UDP/ARP	33.01%
2.b-Z_Eth	IP/DNS/TCP/Raw/UDP/Ether/NTP/ARP/IPv6	0.52%

5.2.1. Training models using all packet layers

In this experiment, we trained all models using all packet layers to understand the contribution of each layer to the model decision (RQ1). As can be seen in Table 9, many of the images had at least one Yellow Hexadecimal Byte residing in the Ethernet layer with a minor combination of other input features. This means that the Ethernet layer had a significant contribution to the classifier. However, the Ethernet layer represents devices' MAC addresses, which are device-dependent, and their addresses do not necessarily relate to the type of network traffic as the same device may generate different kinds of traffic. Further analysis of the effect of other parts of the packet is provided in Section 6. In the next experiments, we test the removal of the Ethernet layer information using the two techniques mentioned previously.

5.2.2. Training models after replacing ethernet with 0s

In this experiment, we test replacing the Ethernet layer with 0s, similar to Garcia et al. (2022) and Li et al. (2022). In the first work, the authors replaced the Ethernet header with 0s to avoid overfitting. Whereas, the latter work randomized the MAC and IP addresses to eliminate their negative effect on feature selection from raw bytes. We proceeded by retraining the models after replacing Ethernet with 0s and evaluated whether we could achieve the same performance on the given task and dataset. To this end, Tables 7 and 8 show that performance metrics maintained similar values despite replacing the Ethernet layer, which strongly supports our hypothesis that classifiers may not necessarily pick the most meaningful features for classification (RQ3).

The obtained results show that, similar to Garcia et al. (2022), the highest activations still occur at some locations of the Ethernet layer despite replacing them with 0s. As shown in Table 10, the percentage of images with at least one Yellow Hex Byte from the Ethernet layer has significantly decreased, yet, the model is still using these empty bytes for classification. With the decreased significance of the Ethernet layer, the model started highlighting more pixels from other layers which are analyzed next.

To maintain brevity, we highlight the most prominent behaviors in Table 11, which are visualized in Fig. 5. In general, we observe that the consistency of contributing network layers varies across the different traffic types from the datasets used. While some network features like IP and TCP show relatively consistent contributions, others like Ethernet and Raw exhibit variability. In the next paragraphs, we use the heatmaps generated with Grad-CAM on the Testing sets of the experiments and explain the most prominently observed behaviors in details. We will be referring to traffic types with matching patterns from Table 12 using their row number.

Layer consistency. For some traffic types, it is observed that the percentages of some layers such as IP and TCP are high enough to assume their relatively consistent contribution in W_Eth and Z_Eth.

This behavior is observed in rows #3, #4, and #16. In particular, IP and TCP contributions are generally high for malicious traffic. In cases like Mal_SQLI, the IP contribution is 96.94% (W_Eth) vs. 98.31% (Z_Eth), showing a high contribution. The same behavior applies to TCP contributions indicating consistent influence. These protocols are foundational to most internet communications, whether benign or malicious.

Layer inconsistency. Some traffic types have inconsistent presence of highly contributing layers. Specifically, no dominant layer is consistently influential in the respective traffic type. For instance, despite being consistent for malicious traffic, IP contributions in application traffic are relatively lower for some traffic types like "Facetime" and "VoIP", suggesting these traffic types rely more on other layers (such as UDP). This behavior is observed across many traffic types in rows #7–8, #10, #12, #14, #27–29, #32 and #38. In particular, VPN traffic demonstrates notable variability in IP and lower layers, likely because of encryption and tunneling. Other layers, such as Ethernet in application traffic like "BitTorrent", and UDP in "Facetime", are high in W_Eth but lower in Z_Eth, indicating notable inconsistencies. For the sake of simplicity, we show the results for IP layer only in Table 11.

Appearance of Ethernet layer. In some cases, the Ethernet layer surprisingly appears only after replacement with 0s. The appearance of the Ethernet layer with varying contributions for traffic type "VPN VoIP" in row #38 in the Z_Eth setting suggests a shift in the relative importance of the Ethernet layer in this traffic type. This is concerning since the Ethernet layer should not be picked especially when it contains empty 0 bytes.

Disappearance of Ethernet layer. The disappearance of the Ethernet layer for some traffic types signifies a shift towards the contributions of other relevant network layers. The only traffic type with this effect was "Weibo" albeit disappearing from an already small margin, where bytes from the Ethernet layer were no longer deemed important in the Z_Eth setting. This was in fact the anticipated and intended behavior for all traffic types after Ethernet replacement.

Absence of Ethernet layer. In some traffic types, the Ethernet layer was absent from both W_Eth and Z_Eth cases. Such behavior is observed in the traffic types in rows #33–37 and indicates that this layer has no impact on them. For instance, the Ethernet layer does not contribute to the traffic type VPN Email (row #33) classification. In this case, other layers such as Raw, TCP and DNS are essential for this traffic type.

Persistence of Ethernet layer. Some traffic types were significantly influenced by the Ethernet layer even after Z_Eth. This persistence is found to be more apparent in the traffic type "WorldOfWarcraft" in row #16, where Ethernet contribution was at a low percentage of 18.97% in W_Eth, and increased to 75.9% in Z_Eth. Similar behavior was observed in rows #9, #13, #29, #31, and #38. As such, this behavior is problematic in network traffic classification and reveals a vulnerability that can be easily exploited by adversarial methods.

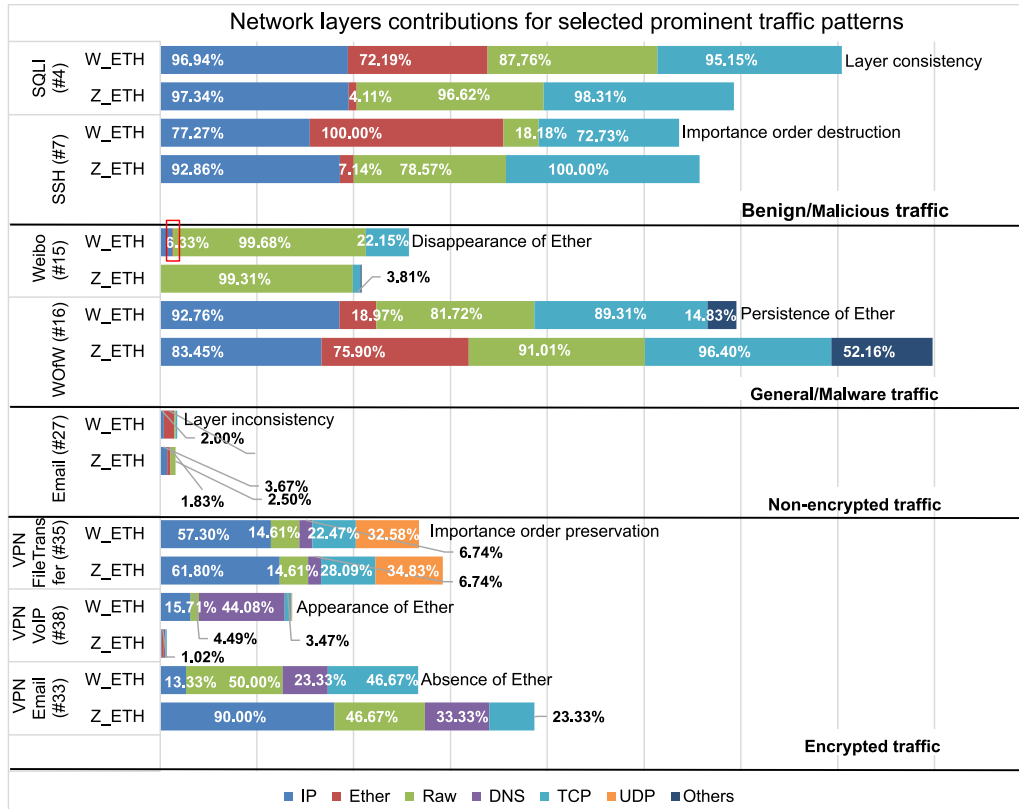


Fig. 5. Visualization of the contributions of traffic patterns from Table 11 grouped by associated datasets. The percentages displayed represent the contributions of designated network layers for each traffic type. We omit the data labels having lower percentages (below 1%) to improve readability. For traffic type #15, we highlight the Ethernet layer contribution in red.

Table 11

Prominent pattern behaviors of most contributing network layers by traffic type. The traffic type ID corresponds to the picked example from Table 12. Values between parenthesis are in percentages, and Ether stands for Ethernet layer.

Pattern	Traffic type ID	Number of matching rows	Traffic type	W_Eth	Z_Eth
Layer consistency	#4	3	Mal_SQLI	IP(96.94), TCP(95.15), Raw(87.76), Ether(72.19)	TCP(98.31), IP(97.34), Raw(96.62), Ether(4.11)
Layer inconsistency (Only for IP layer)	#27	10	Email	Ether(5.33), UPD(3), IP(2), TCP(0.83), Raw(0.67)	IP(3.67), Raw(2.50), Ether(1.83), IPv6(0.33), ARP(0.17)
Appearance of Ether	#38	1	VPN VoIP	DNS(44.08), IP(15.71), Raw(4.49), TCP(3.47), UDP(0.61)	Ether(1.02), DNS(0.82), IP(0.82), TCP(0.61), Raw(0.20)
Disappearance of Ether	#15	1	Weibo	Raw(99.68), TCP(22.15), IP(6.33), Ether(0.32)	Raw(99.31), TCP(3.81), Pad.(0.69), IP(0.35)
Absence of Ether	#33	5	VPN Email	Raw(50), TCP(46.67), DNS(23.33), IP(13.33)	IP(90), Raw(46.67), DNS(33.33), TCP(23.33)
Persistence of Ether	#16	6	WorldOfWarcraft	IP(92.76), Ether(89.31), Raw(81.72), TCP(18.97), Pad.(14.83)	TCP(96.4), Raw(91.01), IP(83.45), Ether(75.9), Pad.(52.16)
Importance order preservation	#35	6	VPN FileTransfer	IP(57.30), UDP(32.58), TCP(22.47), Raw(14.61), DNS(6.74)	IP(61.80), UDP(34.83), TCP(28.09), Raw(14.61), DNS(6.74)
Importance order destruction	#7	32	Mal_SSH	Ether(100), IP(77.27), TCP(72.73), Raw(18.18)	TCP(100), IP(92.86), Raw(78.57), Ether(7.14)

Importance order preservation. The preservation of the importance order of different layers means that each layer maintains its order of influence in both W_Eth and Z_Eth settings. It is observed that a few layers' contributions preserved their importance order, particularly in rows #3, #8, #22, #30, #35, and #36. However, only rows #35 and #22 preserved the exact layers and order.

Importance order destruction. Unlike the previous pattern, for many traffic types, the order of layer importance was changed after undergoing W_Eth process. This behavior frequently occurred with a high number of matching rows (32 samples).

In the next experiment, we retrain the model once again, after completely removing the Ethernet layer from the sessions.

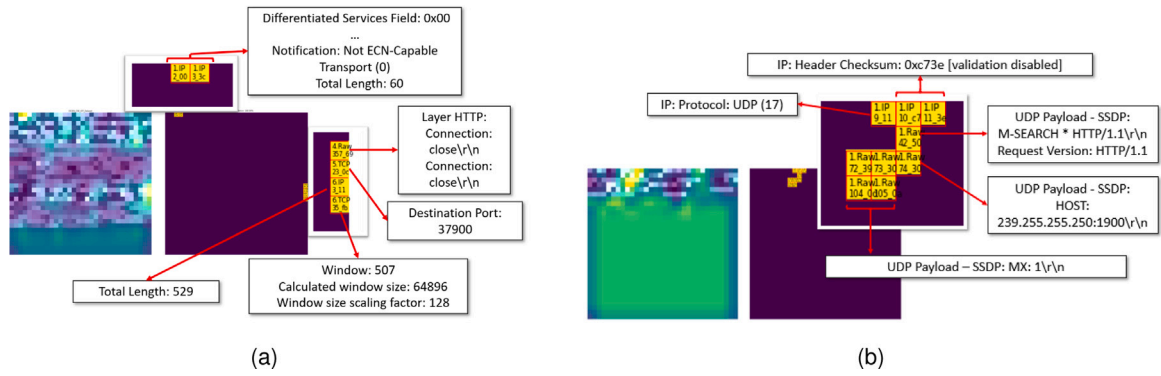


Fig. 6. Visual explanation of decision impacting regions for an attack session (Malicious-SQLI). (a) Correctly classified as Malicious, with a confidence score of 100%. (b) Missclassified prediction as Benign, with a confidence score of 100%.

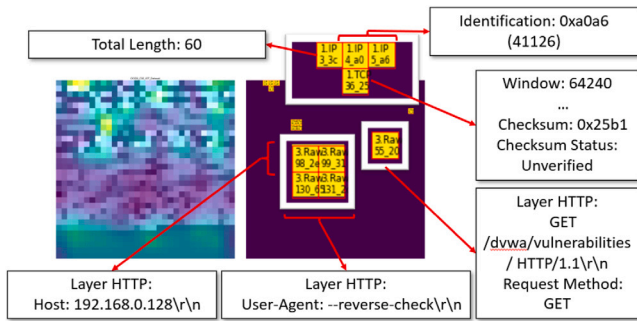


Fig. 7. XSS malicious session correctly classified with a confidence score of 100%.

5.2.3. Training models without ethernet

In an attempt to veer the classifier from using the empty bytes as observed in the previous experiment, we remove the Ethernet layer entirely from the datasets and retrain the models as done in Garcia et al. (2022). As shown in Table 7, the classification performance remains competitively good. However, from Table 8 we observe that in the non-encrypted traffic, the version with No_Eth suffered performance degradation. On the other hand, performance in encrypted traffic improved albeit slightly. We discuss these findings in the next section. From this point onwards, all further experiments will be conducted on the No_Eth variant.

5.2.4. Visualization of SQLI and XSS examples under the Test Case 1.a-No_Eth

To identify which layers/protocols contribute highly to the models' decisions after excluding the Ethernet layer entirely, we use the XAI and Reverse Technique on the test samples from the datasets of each experiment. Due to space constraints, we will show only two samples from malicious classes. The samples are chosen on their significant network layer contributions given in Table 13, and misclassified samples will also be shown to investigate the reason behind the models' decision.

Fig. 6(a) shows the heatmap, binary mask, and data representation of a Malicious SQLI session. The results show that the hexadecimal bytes with the highest contribution to the model's decision are from the IP, TCP, and Raw(HTTP) layers of the session. These bytes map to different details, including the "Total Length" from the IP protocol, the "Destination Port" and "Window Size" from the TCP protocol, and a few bytes representing "Connection: close" from the TCP Payload which is the HTTP layer. From Table 13, we can see that for the MalSQLI class, a large percentage of sessions had highly impacting bytes in the IP layer, more than half the samples had bytes from the Raw layer, and exactly half of them from the TCP layer. As mentioned

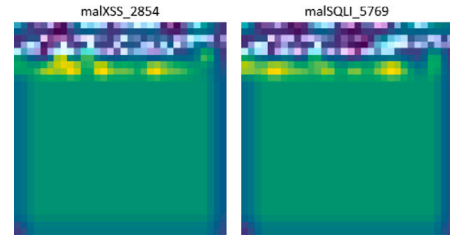


Fig. 8. Misclassified malicious sessions due to trimming of original Pcap file.

in Section 3.3.1, we only look at the bytes with the highest gradients (Yellow) in our work, but that does not mean that the model's decision is solely based on them. Other slightly less important bytes are colored "Light Green" in the heatmap, but for simplicity, we look at the Yellow Hexadecimal Bytes only. Fig. 6(b) shows a misclassification case where a Malicious SQLI session was predicted as Benign, mainly due to the details in the IP and UDP protocols; (i) Layer IP: Header Checksum, and (ii) Details from the UDP Payload. Note that the prediction confidence for the misclassification was 100% which renders the confidence estimate as unreliable. Note that other Benign samples also had highly impacting bytes in the TCP, IP, and UDP protocols, where two of them (IP and UDP highly impacting bytes) are found in the example in Fig. 6(b).

Fig. 8 shows two samples from the Malicious class, XSS and SQLI respectively, that were misclassified by the model due to trimming of the original Pcap file. As mentioned in Section 4.2, the Pcap sessions are trimmed to 1024 bytes to create images of the same size. When the files are trimmed, some Malicious sessions seem to lose their malicious packets. Thus, limiting sessions to 1024 bytes may obfuscate the important Malicious identifiers.

For the XSS example, we can see from Fig. 7 an XSS malicious session correctly classified, with its highly impacting bytes in the IP, TCP and Raw layers. The "Total Length" and "Identification" from the IP layer are some of the highly impacting bytes, while from the TCP layer, the "Window Size" and "Checksum" were found important. From the Raw layer (HTTP), the "Host" address, "User Agent" and the "GET" method with the URL were marked as highly impacting. This demonstrates sensible rationale given that XSS attacks can be identified by looking at the "GET" method. From Table 13, we can see that a high percentage of samples from the BitTorrent class have highly impacting bytes from the IP, TCP, and Raw layers (TCP Payload containing HTTP content), the Raw layer being the highest, at around 93.31% of the samples.

5.2.5. Testing models with images of varying packet numbers

A network traffic classifier operating in a real-world scenario will need to classify traffic early during the session and before it is closed in

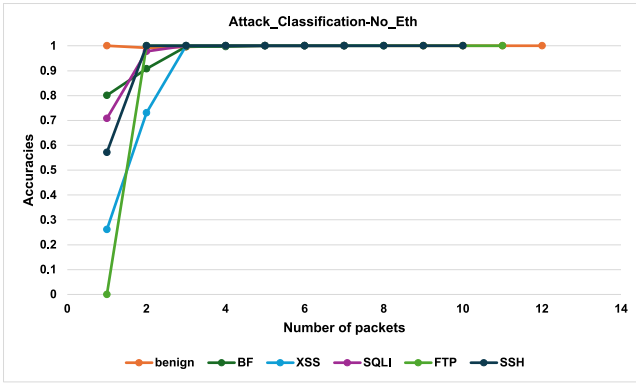


Fig. 9. Accuracies of varying packet numbers in Test Case 1.a-No_Eth.

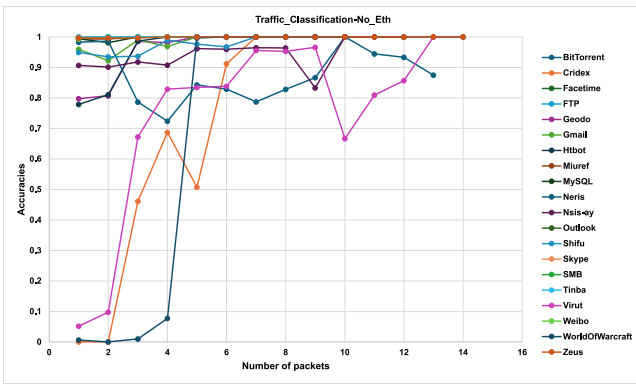


Fig. 10. Accuracies of varying packet numbers in Test Case 1.b-No_Eth.

order to be practical. In the next experiment, we test how the classifier performs when only a few packets of a session are presented to it. To check how early a network traffic classification model can classify a session, we created images with different packet numbers for each session. In other words, if a trimmed session has 5 packets in total, 5 images are created, the first image containing 1 packet, the second containing 2 packets, until the full packet is represented or the image is full. After creating these session variations, we test the models on each image version separately; that is, with the images that have 1 packet only, followed by the images with 2 packets only, ... etc. Tests were conducted for each class separately.

Figs. 9 and 10 show the accuracies of the models on the images with varying numbers of packets in each experiment. Note that the trained classifier is still a binary classifier. All the malicious traffic seen in Fig. 9 are under the Malicious class, but for a more detailed study, we explain the results by traffic type. As can be seen in Fig. 9, the model performed well starting from the third packet in all the traffic. In the case of Benign traffic, the model was able to predict the correct class from the first packet. However, the model misclassified many samples in the case of Malicious traffic with 1 and 2 packets only. This shows that a classifier will perform well enough using as little as the first 3 packets of a session in the considered attack types.

For the Test Case 1.b-No_Eth experiment, the model accuracy was satisfactory starting from the first packet on all traffic types except; Virut, Cridex, and WorldOfWarcraft traffic. Interestingly, for some traffic types, the increase of number of packets led to more misclassified samples. That can be seen in Fig. 10 for Cridex, Nsis-ay, Virut, and Neris traffic. Perhaps, the introduction of the latter packets introduced similarities with other traffic types, causing the confusion.

Concerning the experiments for the Test Case 2.b, Figs. 11 and 12 show the accuracies as a function of session sizes. The model correctly

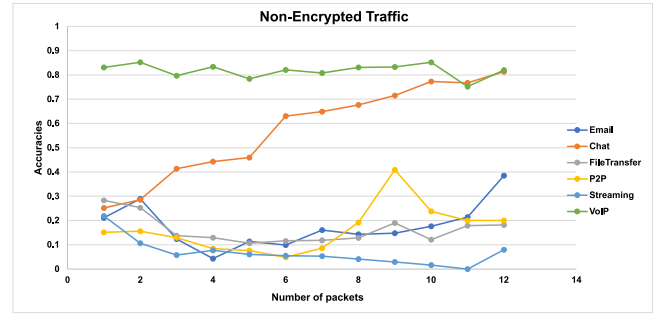


Fig. 11. Performance of the model on non-encrypted traffic as a function of packet count for the Test Case 2.a-No_Eth.

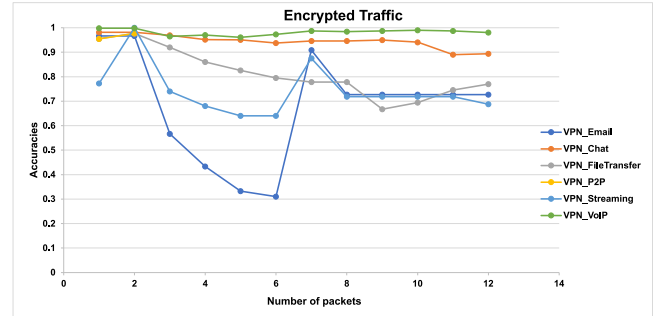


Fig. 12. Performance of the model on encrypted traffic as a function of packet count for the Test Case 2.b-No_Eth.

classified all traffic classes belonging to encrypted traffic from the second packet, as can be seen from Fig. 12. Yet, the model misclassified many samples from the class “VPN_Email” with 3 and 6 packets only. Then, the model recovers its performance when using the first 2 packets of a session only. In contrast, in non-encrypted traffic, the model has poor performance at any stage, except for the classes “VOIP” and “Chat”, as can be seen from Fig. 11. Here, increasing the number of packets does not significantly improve the model performance, which can be a result of highly overlapping patterns.

6. Discussion

Explaining an ML model’s decision aids researchers and practitioners in understanding, trusting, and improving the system. In the case of Network Traffic Classification, it allows experts to understand models’ reasoning and improve their robustness against adversarial attacks. In our work, we used XAI on two main test cases that cover the majority of Network Traffic types. The first Test Case experiments with Benign/Malicious, and General+Malware network sessions in both settings — Binary and Multiclass. The second Test Case studies multi-class classification of Encrypted and Non-Encrypted network sessions.

In the first Test Case (i.e. Attack Traffic Classification), the models looked mainly at the IP layer, TCP layer, and Raw, as can be seen in Table 12 (rows#1-26) and visualized in Fig. 13, Fig. 14 and Fig. 15 for benign and malicious, general traffic and malware traffic, respectively. The exact protocol differed between the classes, however, the features that the models considered are common among the sessions from the same class. For instance, the common highly contributing bytes in the sessions from the SMB class are from the IP layer addresses, the TCP Payload Continuation Data, and the TCP layer port numbers. For the case of encrypted traffic (i.e. Encrypted Traffic Classification), the models classify the images using a combination of features from IP, DNS and TCP layers, as can be seen in Table 12 (rows#27-38) and visualized in Fig. 16 and Fig. 17 for non-encrypted, and encrypted traffic,

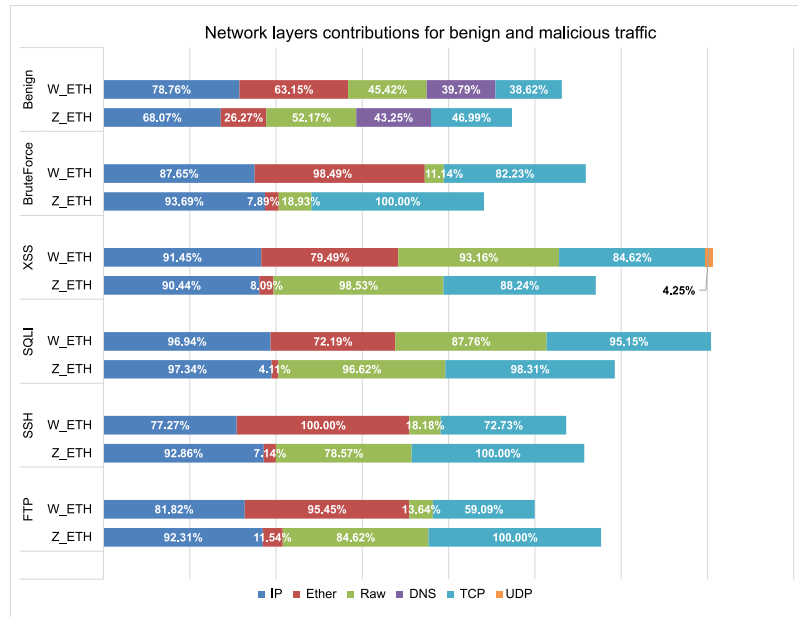


Fig. 13. Visualization of network layers contributions by traffic type (rows#1–6 from Table 12).

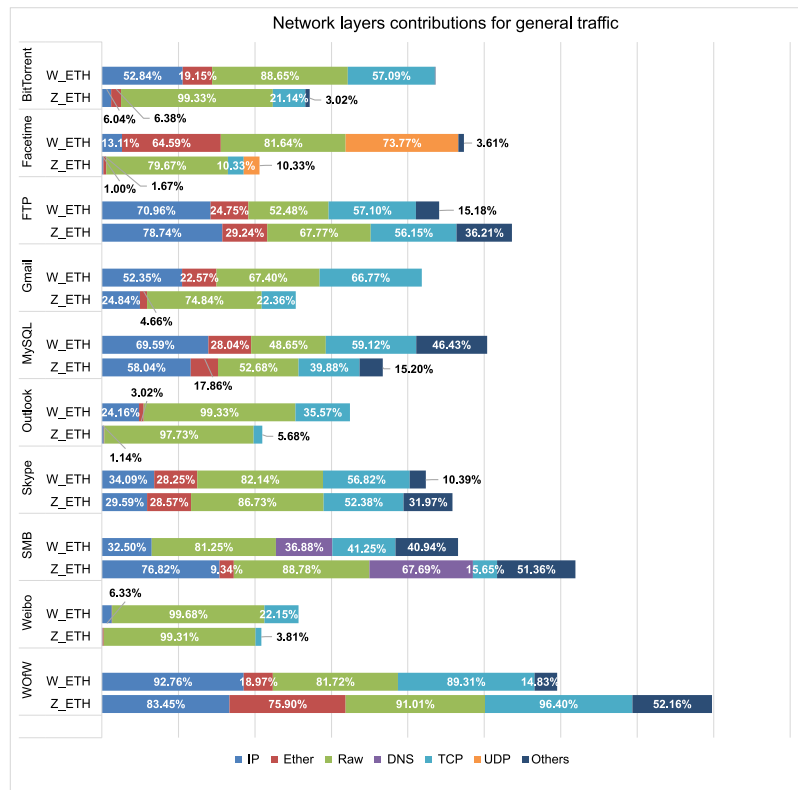


Fig. 14. Visualization of network layers contributions by traffic type (rows#7–17 from Table 12).

respectively. This is in line with other research works like (Mustafa Al-Fayoumi, 2022), where they found out that models mainly relied on the IP packet header. In terms of considered features, the models used common features across the sessions from the same class. For example, for the class VPN VoIP the model used the features like ‘Name’, ‘Name Length’, and ‘Label Count’ as the most important features from DNS layer. From the TCP layer, some of the important features include “TLSv1 Record Layer” and “Handshake Protocol”, which are plain text (Akbari et al., 2021) (i.e. unencrypted data). Our findings are

in line with the findings of other authors (Cherepanov et al., 2022; Elmaghaby et al., 2024). The findings here sufficiently addressed RQ1.

Since we studied the XAI of the models on the images that are most representative of the traffic types, as such these images explain the common features for each network traffic type, we thereby admit that the models in both test cases learned to correctly classify the sessions consistently. However, for the case of misclassification, the models looked in certain cases at different feature combinations. Fig. 6(b) depicts the attributions for an “SQLi” session misclassified as a

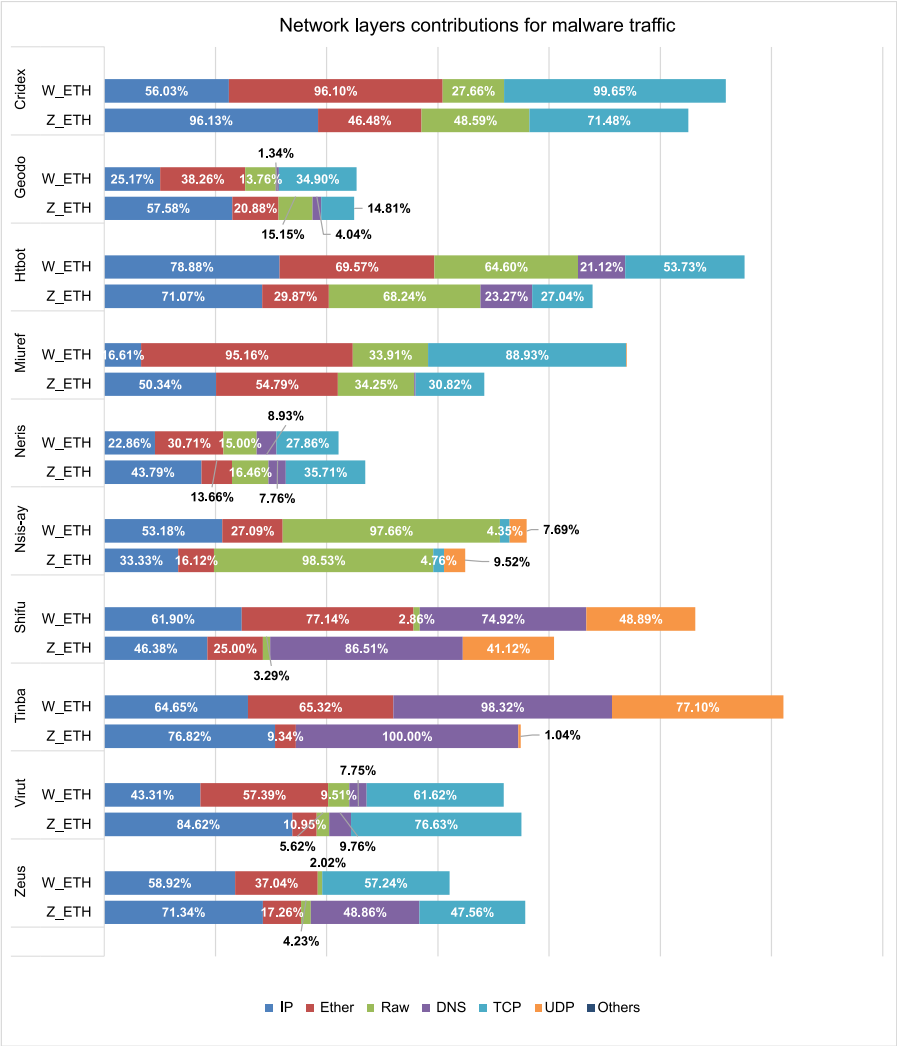


Fig. 15. Visualization of network layers contributions by traffic type (rows#18–26 from Table 12).

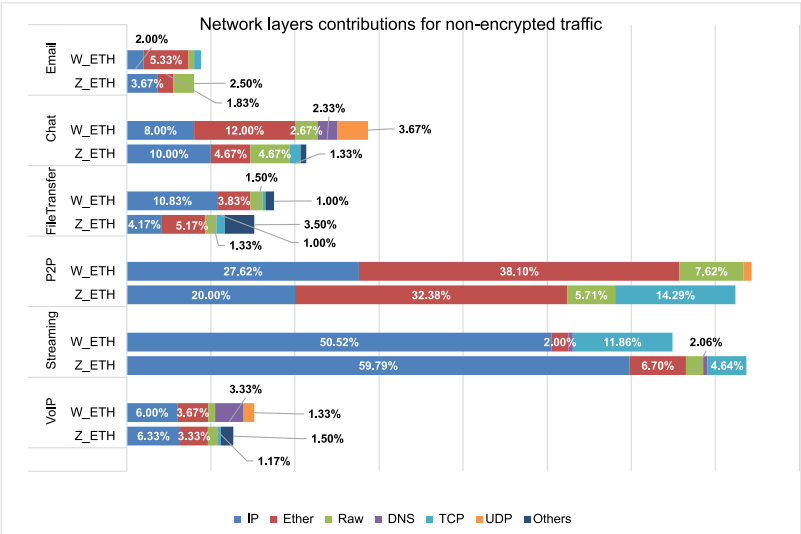


Fig. 16. Visualization of network layers contributions by traffic type (rows#27–32 from Table 12).

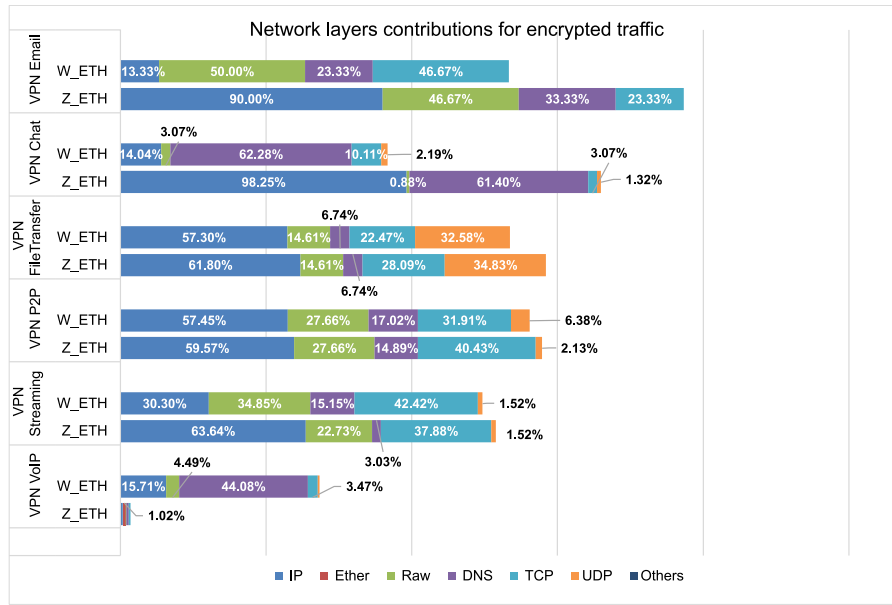


Fig. 17. Visualization of network layers contributions by traffic type (rows#33–38 from Table 12).

“Benign” session, where the highest activations occur in the IP layer and UDP Payload. This answers the RQ2.

Our first concern regarding the meaningfulness of the features used by the model is the use of the features from the Ethernet layer. Such features are not relevant for network traffic tasks since they can change at any period of time as mentioned in De Lucia et al. (2021). Even after masking this layer with 0s, the models still classify the images based on the Ethernet layer. Another encountered issue was that a great portion of the images have been classified or misclassified based on the 0s padded at the end of sessions. An example of such is shown in Fig. 8. This effect is observed in most short sessions. Here, the models learned to classify based on the padded 0s. Perhaps, the padding was more associated with certain traffic types than others. Despite its prevalence, this issue has not been investigated well by previous works yet. We attest that classifiers may indeed consider irrelevant features for network traffic classification, which addresses RQ3.

When using raw network traffic, such unstructured raw data must be trimmed and padded to a fixed-size representation as required by the image-based models used. After investigating XAI and the reversal techniques, we identified two issues associated with the trimming and padding of network packets. On the one hand, when faced with longer network packets, trimming causes data loss (De Lucia et al., 2021) and relevant parts of long sessions were cut off. On the other hand, models unexpectedly used the padding in the images as key features as previously mentioned. For instance, in short sessions having 0s as padding, as can be seen in Fig. 8, the models gave higher importance to those features compared to other meaningful ones. This behavior can potentially be exploited by an adversary to bypass the NIDS by crafting packets with special padding patterns.

Moreover, raw data can be large and can contain a lot of noise and irrelevant information, which was difficult for models to discern. For example, in the case of Encrypted traffic, the models unexpectedly used features from the encrypted TCP payload in combination with others to classify the traffic. Interestingly, when classifying encrypted data, the three classes – Chat, File Transfer, and VoIP – had the impacting bytes from the payload. Whether the encrypted parts of the payload are indeed relevant can be a promising future research direction. Note that the observation of models making decisions on the encrypted parts of the packets was also reported by Lotfollahi et al. (2017). In our case, the impacting bytes were associated with both the header and the payload as reported in Cherepanov et al. (2022). As a result of

this investigation, we assert that when faced with raw data, models may choose unexpected and sometimes encrypted features which raises security concerns (RQ4).

The use of raw network sessions significantly increases data storage and bandwidth requirements (De Lucia et al., 2021). Thus, it may not be feasible to use raw network packets or sessions as features in a real-time traffic classification, which is the core component of many emerging QoS-enabled products and automated QoS architectures. Network traffic classification should ideally be able to identify traffic types or attacks early during the session. Hence, we studied the models’ accuracies while varying the size of raw network traffic. Fig. 11 highlights an unexpected finding in which encrypted traffic sessions achieve high classification accuracy with as few as two packets. This suggests that early detection in resource-constrained environments is viable even for encrypted communication, possibly due to consistent header patterns in encrypted protocols. It is worth mentioning that Grad-CAM was also shown to be suitable for real-time applications in previous studies (Chen et al., 2020). Thus, the proposed method could potentially enable real-time traffic monitoring interpretability analyses, such as observing the network layers being considered for particular communication sessions in real-time, thereby providing valuable insights for network security practitioners. These findings target RQ5.

It is important to note that access to XAI and reverse lookup methods such as the ones proposed in this work has its risks. Attackers may use such techniques to learn what an NIDS looks at during the classification of network traffic, and modify malicious network packets accordingly to evade detection. While we envision these tools to be used in a closed environment during model development to make white-box access to the NIDS model difficult, different risk mitigation techniques may be proposed in future work, such as exploring selective explainable techniques (Wang et al., 2019).

7. Conclusion

In this paper, we leveraged XAI and reverse lookup techniques as explanation methods to understand “Black-box” image-based models in the context of network traffic and attack classification. In particular, we focused on session-based traffic and attack classification. The performance of our models showed strong results on all tested datasets, even in encrypted traffic. An analysis of the highly impacting parts of network sessions was conducted using Grad-CAM to generate activation

Table 12

Contributions of network layers by traffic type (given values are in percentages). Each row represents a specific traffic type and shows the percentage contributions of different network layers under two different feature variations namely W_Eth and Z_Eth, indicating which layers are most prevalent in that type of traffic.

ID	Traffic type	W_Eth	Z_Eth
1	Benign	IP(78.76), Ether(63.15), Raw(45.42), DNS(39.79), TCP(38.62)	IP(68.07), Raw(52.17), TCP(46.99), DNS(43.25), Ether(26.27)
2	Mal_BruteForce	Ether(98.49), IP(87.65), TCP(82.23), Raw(11.14)	TCP(100), IP(93.69), Raw(18.93), Ether(7.89)
3	Mal_XSS	Raw(93.16), IP(91.45), TCP(84.62), Ether(79.49), UDP(4.27)	Raw(98.53), IP(90.44), TCP(88.24), Ether(8.09)
4	Mal_SQLI	IP(96.94), TCP(95.15), Raw(87.76), Ether(72.19)	TCP(98.31), IP(97.34), Raw(96.62), Ether(4.11)
5	Mal_SSH	Ether(100), IP(77.27), TCP(72.73), Raw(18.18)	TCP(100), IP(92.86), Raw(78.57), Ether(7.14)
6	Mal_FTP	Ether(95.45), IP(81.82), TCP(59.09), Raw(13.64)	TCP(100), IP(92.31), Raw(84.62), Ether(11.54)
7	BitTorrent	Raw(88.65), TCP(57.09), IP(52.84), Ether(19.15), Pad.(0.35)	Raw(99.33), TCP(21.14), Ether(6.38), IP(6.04), Pad.(3.02)
8	Facetime	Raw(81.64), UDP(73.77), Ether(64.59), IP(13.11), Pad.(3.61)	Raw(79.67), UDP(10.33), Ether(1.67), IP(1)
9	FTP	IP(70.96), TCP(57.10), Raw(52.48), Ether(24.75), Pad.(15.18)	IP(78.74), Raw(67.77), TCP(56.15), Pad.(36.21), Ether(29.24)
10	Gmail	Raw(67.40), TCP(66.77), IP(52.35), Ether(22.57), Pad.(5.64)	Raw(74.84), IP(24.84), TCP(22.36), Pad.(20.5), Ether(4.66)
11	MySQL	IP(69.59), TCP(59.12), Raw(48.65), Ether(28.04), Pad.(15.20)	IP(58.04), Raw(52.68), Pad.(46.43), TCP(39.88), Ether(17.86)
12	Outlook	Raw(99.33), TCP(35.57), IP(24.16), Ether(3.02)	Raw(97.73), TCP(5.68), Pad.(1.89), IP(1.14), Ether(0.38)
13	Skype	Raw(82.14), TCP(56.82), IP(34.09), Ether(28.25), Pad.(10.39)	Raw(86.73), TCP(52.38), Pad.(31.97), IP(29.59), Ether(28.57)
14	SMB	Raw(81.25), SMB_NPRHG(45), TCP(41.25), NBT_Session(36.88), IP(32.50)	Raw(88.78), NBTSession(67.69), SMB_NPRHG(66.67), SMB_H(19.73), TCP(15.65)
15	Weibo	Raw(99.68), TCP(22.15), IP(6.33), Ether(0.32)	Raw(99.31), TCP(3.81), Pad.(0.69), IP(0.35)
16	WorldOfWarcraft	IP(92.76), TCP(89.31), Raw(81.72), Ether(18.97), Pad.(14.83)	TCP(96.4), Raw(91.01), IP(83.45), Ether(75.9), Pad.(52.16)
17	Cridex	TCP(99.65), Ether(96.10), IP(56.03), Raw(27.66)	IP(96.13), TCP(71.48), Raw(48.59), Ether(46.48)
18	Geodo	Ether(38.26), TCP(34.90), IP(25.17), Raw(13.76), DNS(1.34)	IP(57.58), Ether(20.88), Raw(15.15), TCP(14.81), DNS(4.04)
19	Htbot	IP(78.88), Ether(69.57), Raw(64.60), TCP(53.73), DNS(21.12)	IP(71.07), Raw(68.24), Ether(29.87), TCP(27.04), DNS(23.27)
20	Miuref	Ether(95.16), TCP(88.93), Raw(33.91), IP(16.61), UDP(0.35)	Ether(54.79), IP(50.34), Raw(34.25), TCP(30.82), DNS(0.68)
21	Neris	Ether(30.71), TCP(27.86), IP(22.86), Raw(15), DNS(8.93)	IP(43.79), TCP(35.71), Raw(16.46), Ether(13.66), DNS(7.76)
22	Nsis-ay	Raw(97.66), IP(53.18), Ether(27.09), UDP(7.69), TCP(4.35)	Raw(98.53), IP(33.33), Ether(16.12), UDP(9.52), TCP(4.76)
23	Shifu	Ether(77.14), DNS(74.92), IP(61.9), UDP(48.89), Raw(2.86)	DNS(86.51), IP(46.38), UDP(41.12), Ether(25), Raw(3.29)
24	Tinba	DNS(98.32), UDP(77.1), Ether(65.32), IP(64.65)	DNS(100), IP(76.82), Ether(9.34), UDP(1.04)
25	Virut	TCP(61.62), Ether(57.39), IP(43.31), Raw(9.51), DNS(7.75)	IP(84.62), TCP(76.63), Ether(10.95), DNS(9.76), Raw(5.62)
26	Zeus	IP(58.92), TCP(57.24), Ether and DNS(37.04), Raw(2.02)	IP(71.34), DNS(48.86), TCP(47.56), Ether(17.26), Raw(4.23)
27	Email	Ether(5.33), UPD(3), IP(2), TPC(0.83), Raw(0.67)	IP(3.67), Raw(2.50), Ether(1.83), IPv6(0.33), ARP(0.17)
28	Chat	Ether(12), IP(8), UDP(3.67), Raw(2.67), DNS(2.33)	IP(10), Ether(4.67), Raw(4.67), TCP(1.33), IPv6(0.67)
29	FileTransfer	IP(10.83), Ether(3.83), Raw(1.5), IPv6(1), TCP(0.33)	Ether(5.17), IP(4.17), IPv6(3.50), Raw(1.33), TCP(1)
30	P2P	Ether(38.10), IP(27.62), Raw(7.62)	Ether(32.38), IP(20), TCP(14.29), Raw(5.71), UDP(0.95)

(continued on next page)

Table 12 (continued).

ID	Traffic type	W_Eth	Z_Eth
31	Streaming	IP(50.52), TCP(11.86), Ether(2), DNS(0.52)	IP(59.79), Ether(6.70), TCP(4.64), Raw(2.06), DNS(0.52)
32	VoIP	IP(6), Ether(3.67), DNS(3.33), UDP(1.33), Raw(0.83)	IP(6.33), Ether(3.33), Raw(1.17), IPv6(1.50), TCP(0.33)
33	VPN Email	Raw(50), TCP(46.67), DNS(23.33), IP(13.33)	IP(90), Raw(46.67), DNS(33.33), TCP(23.33)
34	VPN Chat	DNS(62.28), IP(14.04), TCP(10.11), Raw(3.07), UDP(2.19)	IP(98.25), DNS(61.40), TCP(3.07), UDP(1.32), Raw(0.88)
35	VPN FileTransfer	IP(57.30), UDP(32.58), TCP(22.47), Raw(14.61), DNS(6.74)	IP(61.80), UDP(34.83), TCP(28.09), Raw(14.61), DNS(6.74)
36	VPN P2P	IP(57.45), TCP(31.91), DNS(17.02), UDP(6.38)	IP(59.57), TCP(40.43), Raw(27.66), DNS(14.89), UDP(2.13)
37	VPN Streaming	TCP(42.42), Raw(34.85), IP(30.30), DNS(15.15), UDP(1.52)	IP(63.64), TCP(37.88), Raw(22.73), DNS(3.03), UDP(1.52)
38	VPN VoIP	DNS(44.08), IP(15.71), Raw(4.49), TCP(3.47), UDP(0.61)	Ether(1.02), DNS(0.82), IP(0.82), TCP(0.61), Raw(0.20)

Table 13

Contributions of network layers to traffic types utilizing the Ethernet layer, even though zeroed in Z_Eth (values are given in percentages). Each row represents a specific traffic type and details the contributions of different network layers under No_Eth, highlighting the most prevalent layers in each traffic type. NA means that traffic types do not use the Ethernet layer in Z_Eth.

ID	Traffic type	No_Eth
1	Benign	IP(73.22), DNS and Raw(44.99), TCP(38.12), UDP(25.81), Pad.(4.58)
2	Mal_BruteForce	IP(95.89), TCP(95.25), Raw(6.01)
3	Mal_XSS	IP(85.82), TCP(52.99), Raw(37.31)
4	Mal_SQLI	IP(97.58), Raw(52.17), TCP(50)
5	Mal_SSH	IP(100), TCP(95.24), Raw(85.71)
6	Mal_FTP	TCP(100), IP and Raw(91.3)
7	BitTorrent	Raw(93.31), TCP(89.97), IP(72.91)
8	Facetime	Raw(84.43), Pad.(11.98), IP(9.58)
9	FTP	TCP(86.81), Raw(40.63), IP(18.40), UDP(7.64), Pad.(0.69)
10	Gmail	TCP(84.28), Raw(59.87), IP(5.02)
11	MySQL	TCP(85.05), Raw(44.86), IP(14.95), UDP(5.61), Pad.(0.62)
12	Outlook	TCP(92.39), Raw(84.08)
13	Skype	Raw(89.86), TCP(67.39), IP(2.54)
14	SMB	Raw(80.76), TCP(68.73), DNS(29.21), NBT_Session(23.02), SMB_NBRHG(2.75)
15	Weibo	NA
16	WorldOfWarcraft	TCP(98.66), Raw(64.55), IP(51.17)
17	Cridex	TCP(79.39), Raw(49.62)
18	Geodo	TCP(82.12), Raw(17.22), UDP(1.66), Pad.(0.66)
19	Htbot	Raw(57.10), TCP(52.81), UDP(24.75), Pad.(5.28), SMB_H(0.99)
20	Miuref	TCP(64.8), Raw(37.17), UDP(0.66), SMB_H and LLMNRQuery(0.33)
21	Neris	TCP(87.80), Raw(13.9), UDP(6.1), Pad.(2.71), IP(1.36)
22	Nsis-ay	Raw(97), Pad.(68), TCP(8.33), IP and NBNSHeader(0.33)
23	Shifu	UDP(89.55), Pad.(24.48), Raw(4.78), TCP(3.28)
24	Tinba	UDP(98.98), Pad.(27.3)
25	Virut	TCP(90.97), UDP and Raw(7.99), Pad.(4.51), IP(1.39)
26	Zeus	UDP(48.39), TCP(38.71), Pad.(28.71), Raw(2.9), IP(0.65)
27	Email	LLMNRQuery(21.67), IP(13.67), IPv6(18.67), Raw(12.50), UDP(2.50)
28	Chat	IP(76.33), LLMNRQuery(74.67), IPv6(37.33), Raw(19.67), UDP(7.33)
29	FileTransfer	IP(48.50), IPv6(31), Raw(5.67), LLMNRQuery(4.5), UDP(2.67)
30	P2P	Raw(38.10), TCP(27.62), IP(20), UDP(2.86)
31	Streaming	IP(52.58), TCP(40.21), DNS(14.43), Raw(7.22), UDP(3.61)
32	VoIP	IPv6(20.33), IP(19), DNS(18.33), LLMNRQuery(13.83), Raw(4.33)
33	VPN Email	NA
34	VPN Chat	NA
35	VPN FileTransfer	NA
36	VPN P2P	NA
37	VPN Streaming	NA
38	VPN VoIP	DNS(61), IP(1.02), TCP(0.41), UDP(0.20), Raw(0.20)

maps. The regions of high gradients were subsequently mapped to packet information. Consistent highlighting of high-impact pixels was observed across a few traffic types like Mal_SQLI. Moreover, experimental results revealed that for normal traffic, the models often relied on IP addresses, TCP hosts' information, and TCP payloads. Surprisingly, encrypted payloads in some traffic classes, like "File Transfer", were among the impacting bytes.

The developed approach is a practical method to understand the behavior of future image-based network session classifiers. Moreover, the proposed method can help reveal potential vulnerabilities and improve robustness during model development. Additionally, given the efficiency of Grad-CAM and CNN-based models in real-time applications, our method could potentially enable real-time traffic monitoring interpretability analyses, such as observing the network layers being considered for particular communication sessions in real-time. The

time-series information produced by the interpretability module could then be used for more advanced intrusion detection mechanisms.

While the proposed technique can reveal valuable insights about an image-based NIDS behavior, it also has some limitations. For instance, the proposed method uses a preset session size to represent a network session as a fixed size image. While this may encapsulate most sessions, there exist long sessions which may not fit within the allowed size. Furthermore, Grad-CAM provides valuable explanations of factors affecting a model's decision. However, it does not explain complex interactions among the features themselves, and feature extraction and decision-making.

Future research could address these limitations by investigating complementary techniques such as SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017) to better capture the complex interactions between feature extraction and decision-making in CNN/DNN-based NIDS models, providing deeper insights into model behavior and improving model interpretability. Additionally, various other classification models such as Visual Transformers and model enhancements like attention-based mechanisms can also be investigated. The effect of using these techniques on interpretability results is a promising future research direction. Likewise, exploring adversarial robustness in image-based NIDS using simulations and quantitative data is an intuitive next step to leverage the XAI method introduced in this work.

Various data representations, like graph (Zhong et al., 2023) and tabular representations (Chen et al., 2023), can also be explored, while comparing the performance and explainability of models trained on these representations with the image-based model. Future work could also focus on mitigating the challenges related to real-world operation like computational constraints through optimizing explainability algorithms for efficiency, and exploring seamless integration strategies with existing security infrastructures. In particular, hardware-accelerated explainability solutions, such as dedicated ASICs, GPUs, TPUs, etc. can significantly enhance computational efficiency by parallelizing explainability computations.

CRedit authorship contribution statement

Ayah Abdel-Ghani: Visualization, Methodology, Analysis, Writing – original draft, Writing – review & editing. **Jezia Zakraoui:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Abdulaziz Al-Ali:** Funding acquisition, Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Abdelhak Belhi:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Sandy Rahme:** Writing – review & editing. **Abdelaziz Bouras:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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